

	vibrations. This results first in thermal energy, then light being emitted.
4 (a)	Faraday's law of electromagnetic induction states that the induced e.m.f. in a circuit is directly proportional to the rate of change of magnetic flux linkage through the circuit.

4(b)(i)	$\frac{N_s}{N_p} = \frac{V_s}{V_p}$ $N_p = N_s \frac{V_p}{V_s}$ $= (260) \frac{240}{\left(9.0 / \sqrt{2}\right)}$ $= 9805$
4(b)(ii)	The output p.d. and input p.d. are in phase. According to Faraday's law of electromagnetic induction, the e.m.f. induced in the secondary coil (i.e. the output p.d.) is directly proportional to the rate of change of magnetic flux linkage in the core. The rate of change of magnetic flux linkage is in turn dependent on the alternating input voltage. Hence the output p.d. and input p.d. are in phase.
5(a)	$\lambda = \frac{hc}{E}$